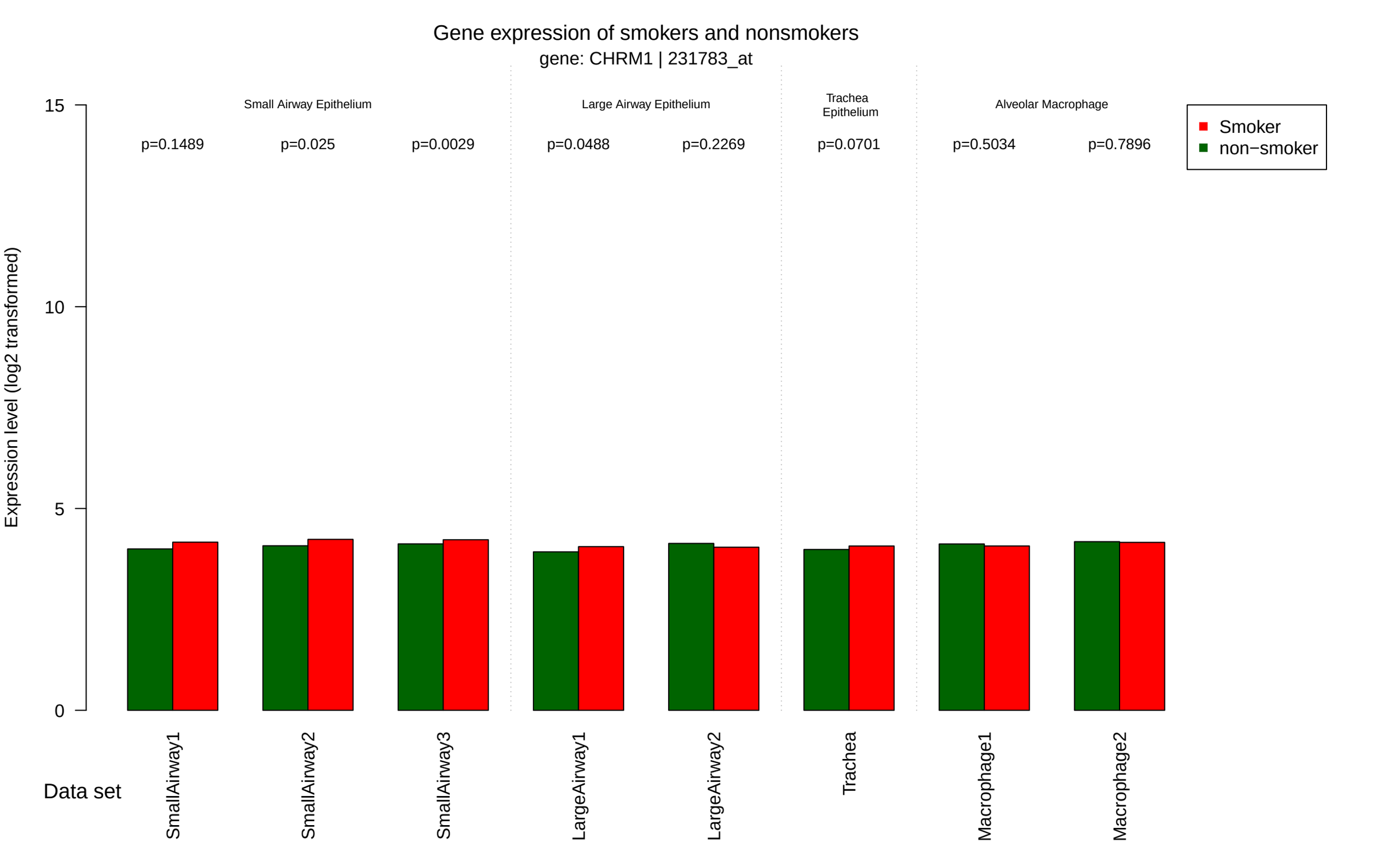
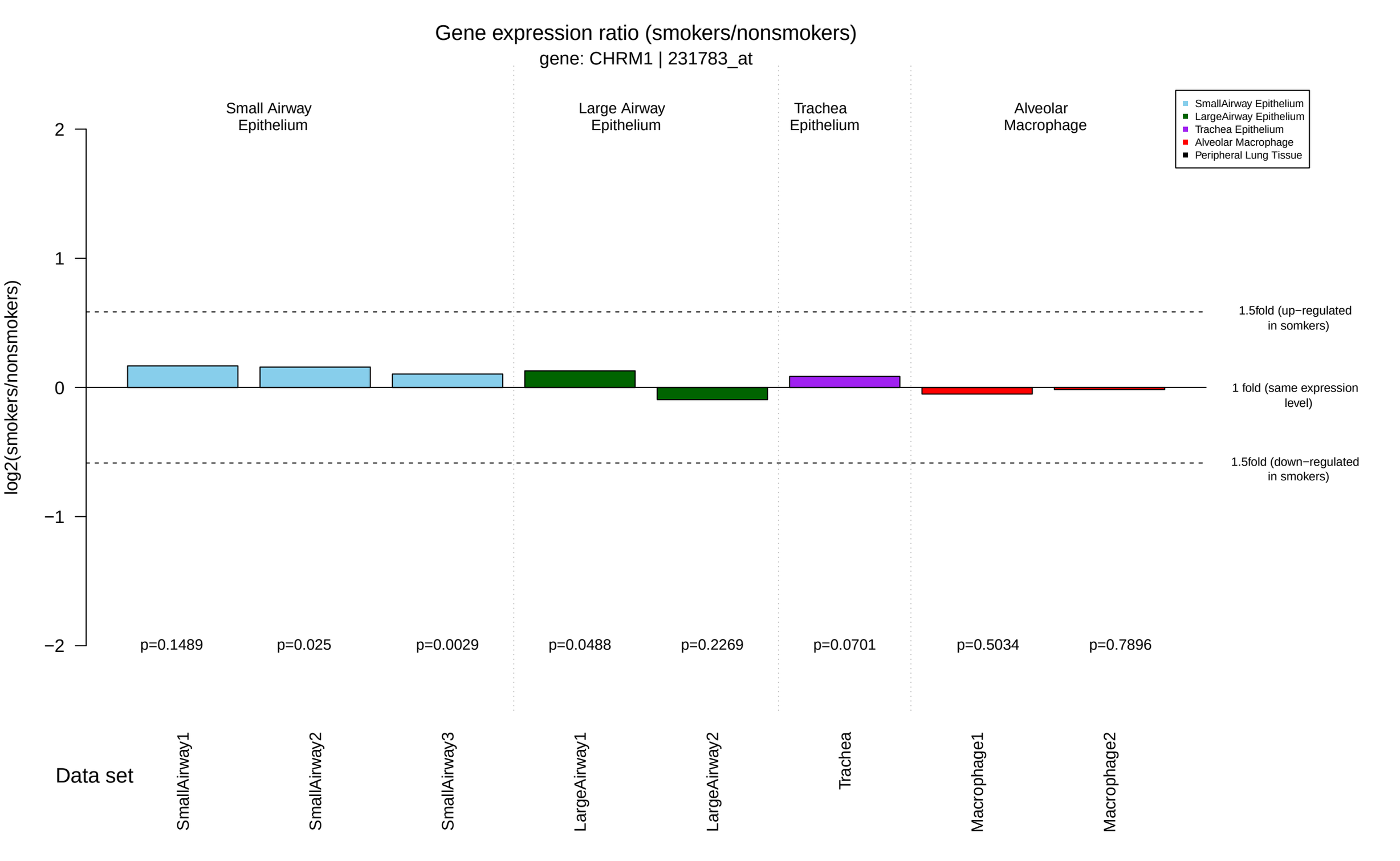
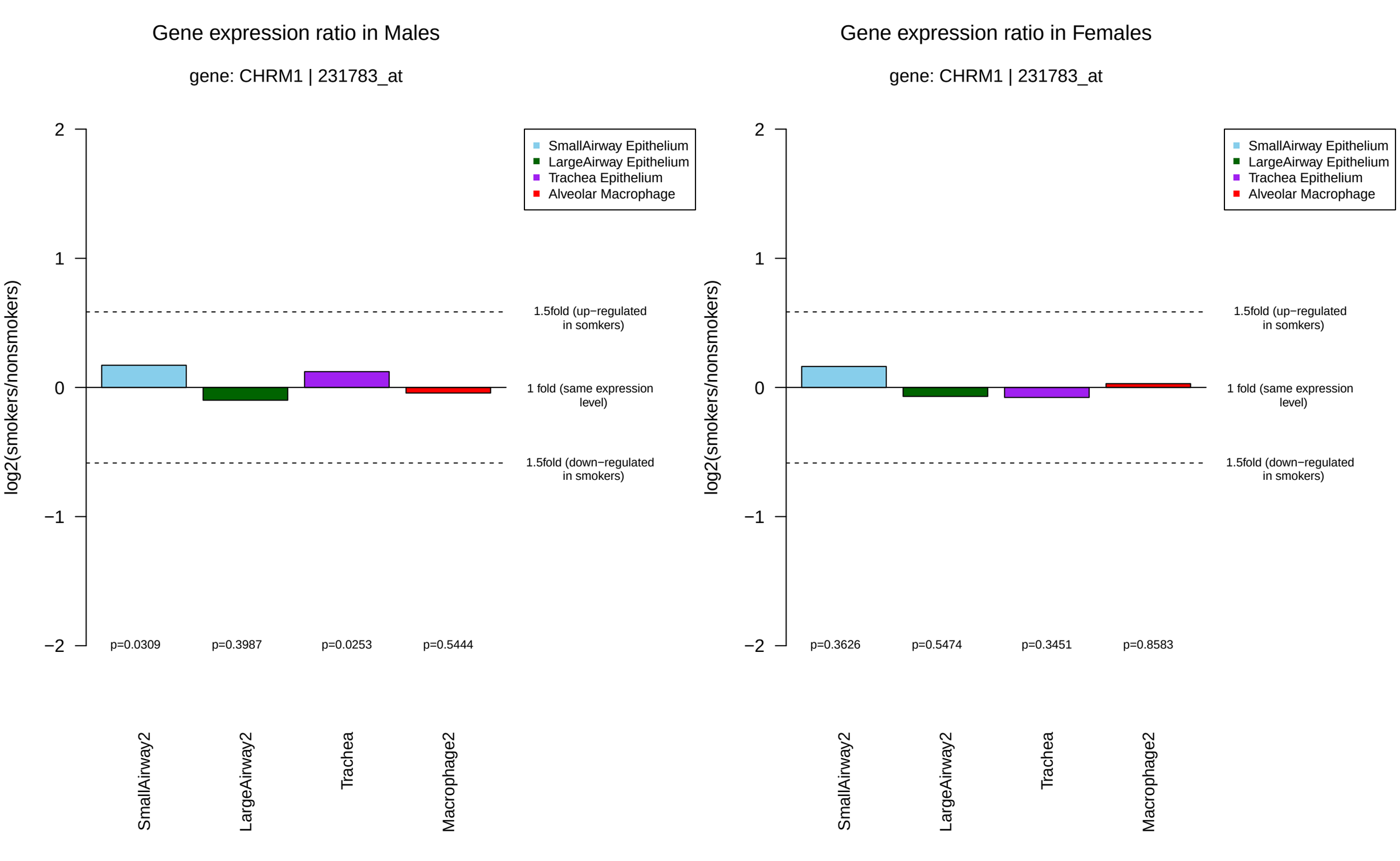
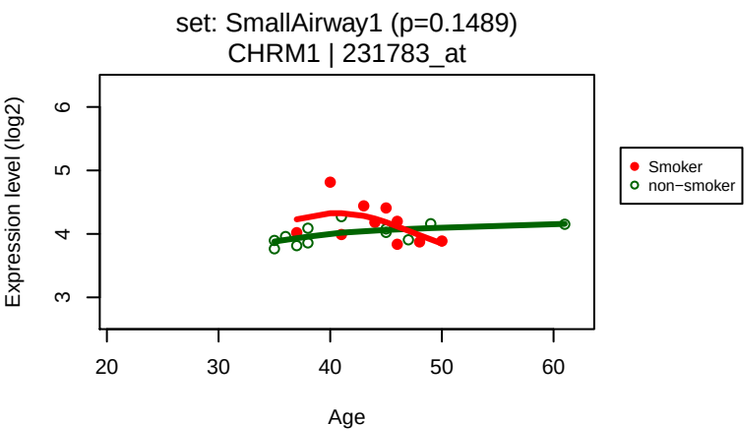
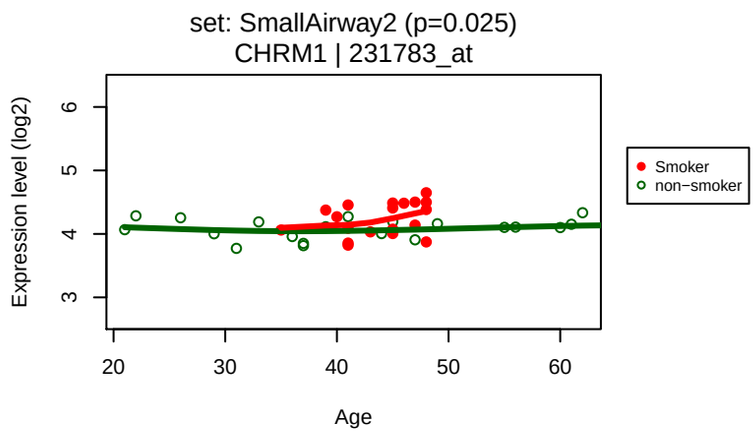
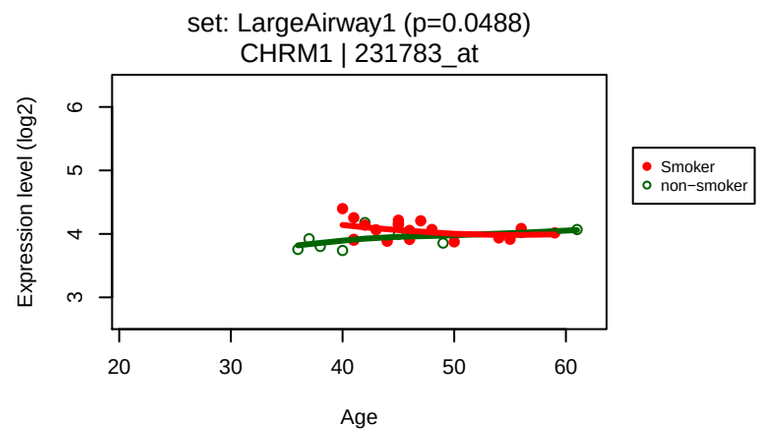

Gene expression of smokers and nonsmokers
gene: CHRM1 | 231783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1489
p=0.025
p=0.0029
p=0.0488
p=0.2269
p=0.0701
p=0.5034
p=0.7896
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

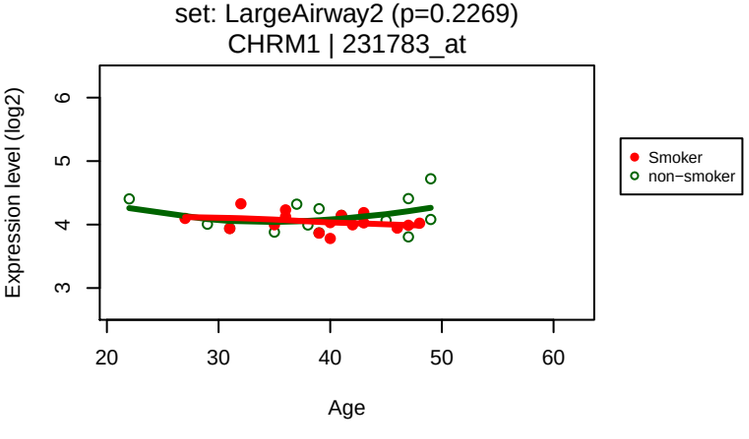

Gene expression ratio (smokers/nonsmokers)
gene: CHRM1 | 231783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1489
p=0.025
p=0.0029
p=0.0488
p=0.2269
p=0.0701
p=0.5034
p=0.7896
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

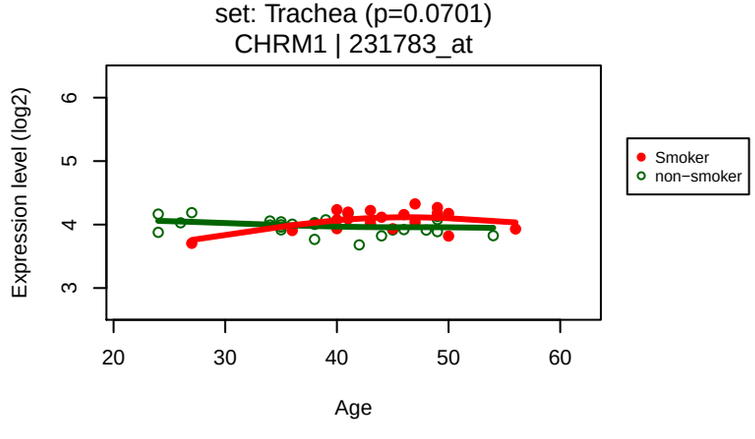

Gene expression ratio in Males
gene: CHRM1 | 231783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0309
p=0.3987
p=0.0253
p=0.5444
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM1 | 231783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3626
p=0.5474
p=0.3451
p=0.8583
SmallAirway2
LargeAirway2
Trachea
Macrophage2

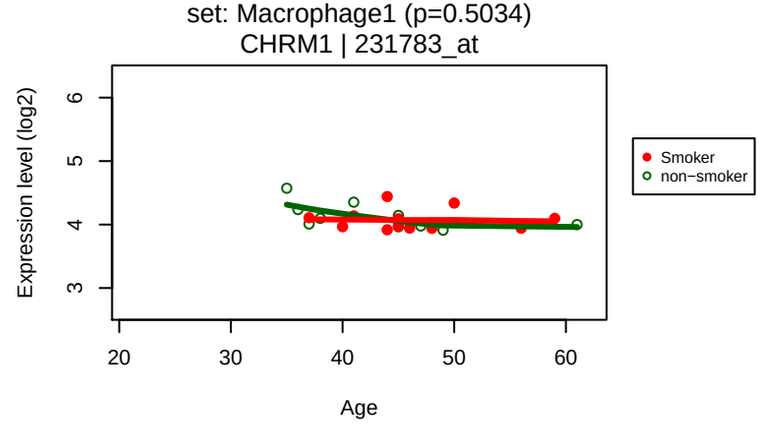

set: SmallAirway1 (p=0.1489)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker

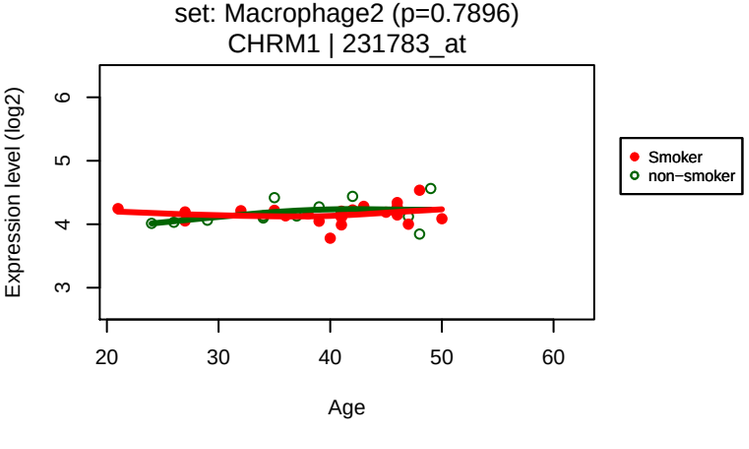

set: SmallAirway2 (p=0.025)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker

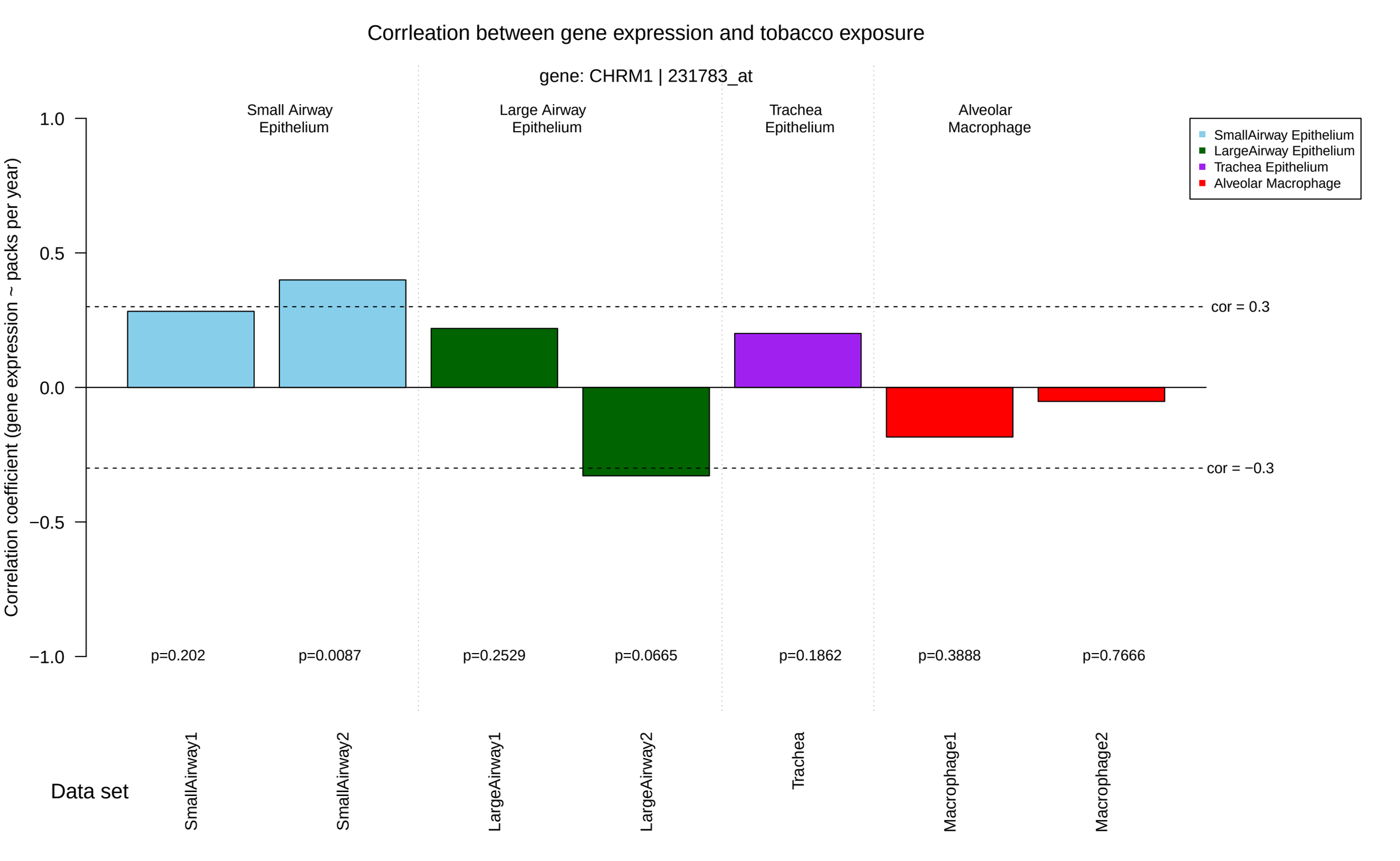

set: LargeAirway1 (p=0.0488)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2269)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0701)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5034)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7896)
CHRM1 | 231783_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM1 | 231783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.202
p=0.0087
p=0.2529
p=0.0665
p=0.1862
p=0.3888
p=0.7666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2